

Ex.No:1: Simulation of FDMA in MATLAB

Objective 1

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;
CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time
(s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel
Bandwidth per User')
```

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```
1  clc; clear; close all;
2  fs=1000; t=0:1/fs:1;
3  TDM=600; Users=3;
4  CDM=TDM/Users;
5  fc=[100 300 500];
6  m=sin(2*pi*10*t);
7  s1=m.*cos(2*pi*fc(1)*t);
8  s2=m.*cos(2*pi*fc(2)*t);
9  s3=m.*cos(2*pi*fc(3)*t);
10 subplot(2,1,1)
11 plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
12 title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
13 subplot(2,1,2)
14 bar(1:Users,CDM*ones(1,Users)),grid on
15 xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

Figure 1 X

The figure consists of two subplots. The top subplot, titled 'FDMA Channel Allocation', shows the amplitude of three sinusoidal signals over time (0 to 1 second). The signals are color-coded: red for the first user, green for the second, and blue for the third. The bottom subplot, titled 'Channel Bandwidth per User', is a bar chart showing that each of the three users is allocated an equal bandwidth of approximately 200 kHz.

User	Channel Bandwidth (kHz)
1	200
2	200
3	200

Objective 2

```
clc; clear; close all;

BW = 200;

Users = 3;

M = 2;

% Channel Bandwidth (kHz)

% BPSK Modulation

DataRate = BW*log2(M);

% Data Rate (kbps)

Rate = DataRate*ones(1,Users);

disp(' User  Data Rate (kbps)')

disp([(1:Users)' Rate'])

figure

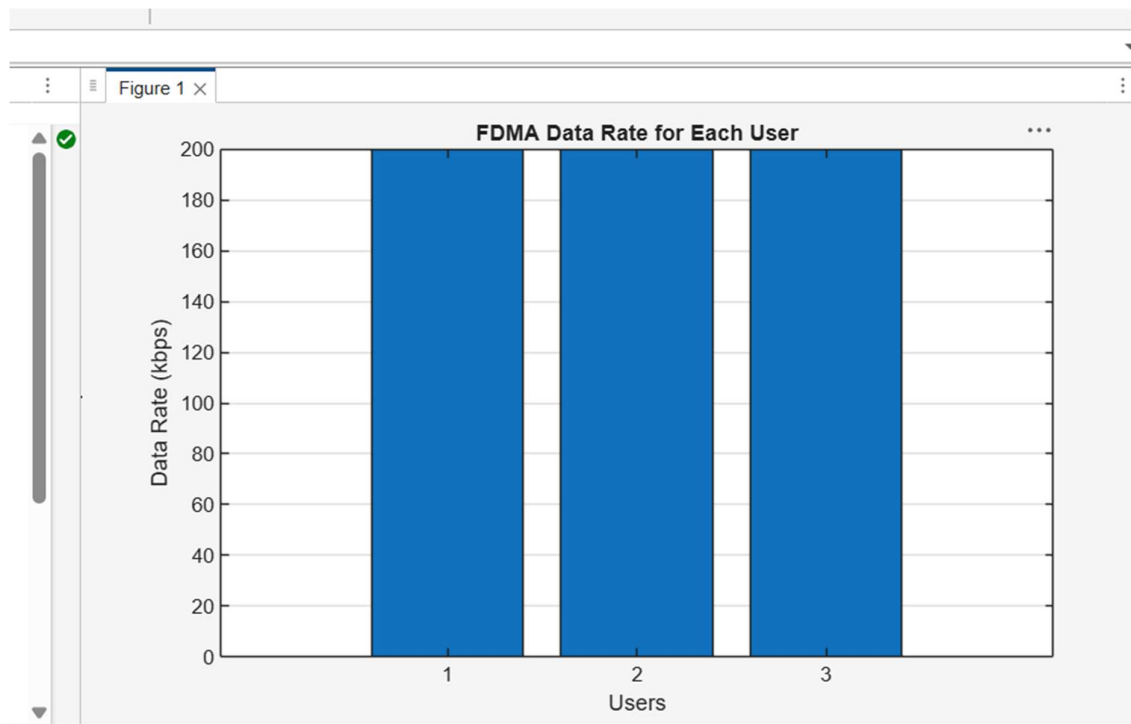
bar(1:Users,Rate)

xlabel('Users')

ylabel('Data Rate (kbps)')

title('FDMA Data Rate for Each User')

grid on
```



Objective 3

```
clc; clear; close all;
```

```
Ps = 1;
```

```
Pn = [0.1 0.05 0.01];
```

```
% Signal Power (W)
```

```
% Noise Power (W)
```

```
Users = 1:3;
```

```
SNR = 10*log10(Ps./Pn);
```

```
% SNR in dB
```

```
disp('User  Noise Power(W)  SNR(dB)')
```

```
disp([Users' Pn' SNR'])
```

```
figure
```

```
bar(Users,SNR)
```

```
xlabel('Users')
```

```
ylabel('SNR (dB)')
```

```
title('Signal-to-Noise Ratio of FDMA Users')
```

```
grid on
```

